

One-bit Single-Measurement Spectral Compressed Sensing via Hankel Matrix Factorization

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Abstract—This paper studies the problem of recovering a spectrally sparse signal from its one-bit single-measurement vector. Most previous methods depend on multiple measurement vectors, which face inherent challenges in real-time critical scenarios. Regarding the single-measurement vector, only a few on-grid approaches have been proposed; however, these techniques suffer from estimation errors due to grid mismatch. To facilitate gridless reconstruction under a one-bit single-measurement vector, a novel optimization model is devised by leveraging both the low-rank property of the lifting Hankel matrix and the sign consistency between the target signal and its estimate. Consequently, a one-bit spectral compressed sensing algorithm via Hankel matrix factorization, termed OSCAR, is proposed by incorporating a one-sided ℓ_p norm smoothing technique. Furthermore, an accelerated OSCAR algorithm, named Acc-OSCAR, is developed by integrating momentum techniques to expedite convergence. Theoretical results characterize the convergence analysis for the proposed algorithms. Numerical results confirm the excellent performance of the proposed algorithms.

Index Terms—Spectral compressed sensing, one-bit, Hankel matrix completion, matrix factorization

I. INTRODUCTION

One-bit spectral compressed sensing aims to recover spectrally sparse signals from one-bit measurements. Owing to its hardware efficiency and energy-saving advantages in analog-to-digital conversion, one-bit spectral compressed sensing has attracted significant attention in applications ranging from automotive radar and underwater sonar to channel estimation [1], [2].

Let $\mathbf{x}^* = [x_0^*, x_1^*, \dots, x_{n-1}^*]^T \in \mathbb{C}^n$ denote the original single-measurement vector for a spectrally sparse signal, whose elements are a superposition of r complex sinusoids

$$x_t^* = \sum_{a=1}^r d_a e^{(i2\pi f_a - \tau_a)t}, \quad (1)$$

where $t = 0, \dots, n-1$, $f_a \in [0, 1)$ is the a -th normalized frequency, d_a is the amplitude of the a -th sinusoids, and τ_a is

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the a -th damping factor. Let $\mathbf{y}$ be its one-bit sampling, defined as

$$\mathbf{y} = \text{csgn}(\mathbf{x}^*) = \sum_{t=1}^n \text{csgn}(x_t^*) \mathbf{e}_t, \quad (2)$$

where $\mathbf{e}_t$ is the canonical basis of $\mathbb{R}^n$, $\text{csgn}(\cdot)$ is defined as

$$\text{csgn}(x) = \text{sgn}(\text{Re}(x)) + j \text{sgn}(\text{Im}(x)), \text{ for } x \in \mathbb{C}, \quad (3)$$

and $\text{sgn}(\cdot)$ is defined as

$$\text{sgn}(h) = \begin{cases} +1, & h \geq 0, \\ -1, & h < 0, \end{cases} \text{ for } h \in \mathbb{R}. \quad (4)$$

Here, we assume that n is odd as [3]; otherwise, we can delete the last entry of $\mathbf{y}$ to force it odd.

By simplifying signals into binary representations, one-bit spectral compressed sensing significantly reduces data volume, thereby lowering sampling and storage costs. Meanwhile, replacing analog-to-digital converters with comparators minimizes hardware costs and power consumption. However, one-bit sampling inevitably results in the loss of both amplitude and phase information, thereby complicating the reconstruction of the original signal. While the exact amplitude of the desired signal cannot be recovered, it is still possible to reconstruct a scaled version of the signal from one-bit measurements, from which the underlying frequencies and damping factors can be estimated. This naturally raises the question: *how can we efficiently and accurately reconstruct a scaled version of the desired signal?*

To recover the scaled desired signal from one-bit measurements, several kinds of strategies have been proposed, including on-grid approaches based on one-bit compressed sensing [4], [5], [6], [7], one-bit atomic norm minimization methods [8], [9], [10], [11], and deep learning-based algorithms [11], [12], [13], [14]. However, most of the current algorithms focus on multiple measurement vectors, which demonstrates inherent limitations in real-time critical scenarios, particularly in high-mobility applications such as autonomous driving systems and unmanned combat platforms. Only a limited number of on-grid algorithms demonstrate applicability in single-measurement vector scenarios, but these methods suffer from performance degradation when subjected to grid mismatch conditions.

A. Contributions

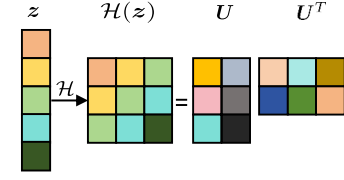
This paper considers achieving gridless spectral compressed sensing under a one-bit single-measurement vector based on low-rank Hankel matrix factorization. The main contributions are as follows:

- 1) A non-convex optimization model is formulated by exploiting the low-rank characteristic of the lifting Hankel matrix, along with ensuring sign consistency between the original and estimated signals, as shown in Fig. 1. The low-rank Hankel matrix factorization is utilized to reduce the dimension of the variables, while the one-sided ℓ_p norm is employed to encourage sign consistency.
- 2) We develop a one-bit single-measurement spectral compressed sensing method via Hankel matrix factorization called OSCAR by utilizing a one-sided ℓ_p norm smoothing technique. Additionally, we present an accelerated variant, Acc-OSCAR, which incorporates momentum strategies to improve convergence performance. Furthermore, to boost computational efficiency, we implement a rapid gradient computation strategy for the proposed algorithms by employing the Fast Fourier Transform (FFT).
- 3) We conduct a theoretical analysis and provide experimental validation for the algorithm's convergence performance in single-measurement vector scenarios, thereby demonstrating the efficacy of the proposed OSCAR methods.

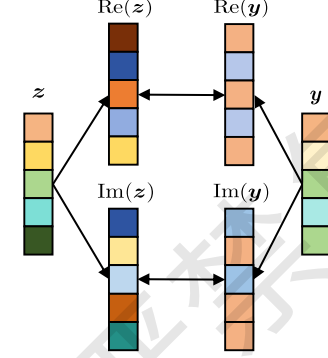
B. Literature Review

Standard spectral compressed sensing considers the reconstruction of spectrally sparse signals from unquantized measurements. There are two main kinds of gridless algorithms for standard spectral compressed sensing, which are the atomic norm minimization (ANM) method and the Hankel matrix completion method. The atomic norm minimization method first designs an atomic norm over a continuous dictionary, then establishes an atomic norm minimization model, and solves it through an equivalent semidefinite programming formulation. For instance, to relax the requirement on spectral separation, Yang and Xie proposed a reweighted atomic norm minimization method that enhanced spatial spectral estimation accuracy under single-measurement conditions [15]. Qian et al. developed a generalized atomic norm-based model, achieving precise recovery of missing data in multiple measurement vectors [16]. The Hankel matrix completion method exploits the low-rank property of the lifting Hankel matrices of the original signal, formulates a low-rank optimization model, and solves it via convex/nonconvex optimization algorithms, which effectively addresses spectral compressed sensing. For example, Chen and Chi proposed a structured matrix completion algorithm that achieves high-probability exact signal reconstruction from a single-measurement vector [17]. Cai et al. studied a projected gradient descent based on asymmetric Hankel factorization and achieved linear convergence [18]. Li et al. introduced a symmetric Hankel structure to the optimization model, which enables exact signal reconstruction with less storage and computational time under a single-measurement vector [3].

Building upon unquantized spectral compressed sensing and one-bit compressed sensing frameworks, researchers have begun to study one-bit spectral compressed sensing, which mainly includes:



(a) Low-rankness of lifting Hankel matrix



(b) Sign Consistency

Fig. 1. Low-rankness of lifting Hankel matrix and sign consistency.

1) One-bit subspace methods via covariance matrix estimation. These approaches first estimate the actual covariance matrix from one-bit data before applying subspace-based spectrum estimation. For instance, Huang and Liao proposed the one-bit multiple signal classification (MUSIC) algorithm [19], which employs a weighted combination of one-bit covariance matrices and identity matrices for covariance estimation, achieving comparable performance to unquantized MUSIC at low Signal-to-noise ratio (SNR). Xiong et al. enhanced estimation accuracy through exchange matrix operations for covariance reconstruction under low SNR conditions [20]. However, these methods demonstrate significant performance gaps compared to unquantized subspace approaches at high SNR. Besides, they are limited to multiple measurement vectors, failing in the case of single-measurement vectors.

2) On-grid methods via one-bit compressed sensing. These techniques construct grid-based steering matrices and employ one-bit compressed sensing algorithms for spectral estimation. Stöckle et al. proposed a complex binary iterative hard-hard-thresholding algorithm with low computational complexity [4]. Wang et al. developed a grid-based approach that incorporates basis update and backtracking strategies and refined threshold functions to address off-grid issues [5]. Chen et al. solved a maximum likelihood estimator through conjugate gradient and steepest descent methods [6]. Li et al. carried out dual-sparsity optimization with proximal alternating minimization for sign-flip correction [7]. While current grid-based methods handle both single- and multi-measurement vectors, their estimation accuracy remains limited by grid mismatch errors.

3) One-bit atomic norm minimization approaches. These methods formulate convex optimization problems through sign consistency and atomic norm constraints, subsequently converted to semidefinite programming for scaled signal recovery. Tang et al. solved binary atomic norm minimization using

accelerated proximal gradient [8]. Zheng and Yang exploited Hankel-Toeplitz structures to conduct reweighted atomic norm minimization for multiple measurement vectors [9]. Li et al. carried out log-determinant-based approximate atomic norm minimization with noise constraints [10]. Feng et al. combined maximum a posteriori estimation with joint basis-sparse signal optimization in an off-grid iterative reweighting algorithm [11]. Despite their high estimation accuracy, these convex optimization methods suffer from prohibitive computational complexity, which limits their practical implementation.

4) Deep learning-driven one-bit algorithms. Emerging deep learning approaches aim to enhance computational efficiency, though research remains in its infancy. Xiao et al. developed a deep unfolding model for a single-measurement vector by integrating grid-based methods with sign consistency, improving mid-to-high SNR performance [11]. Ji et al. proposed a Transformer-enhanced subspace network for multiple measurement vectors, outperforming conventional subspace methods under both low-bit and one-bit quantization conditions [12]. Hu et al. proposed a learning-based sparse Bayesian framework that integrates unrolled neural networks to achieve high-precision off-grid estimation from a single-measurement vector [14]. However, these algorithms rely on a large amount of training data and usually require fixing the dimension of the input signals.

C. Organization and Notations

Organization. The paper is organized as follows. Section II provides the problem formulation by exploring the low-rankness of the lifting Hankel matrix and the sign consistency between the original signal and the estimated signal. Section III proposes the smooth approximation of the objective function. Section IV presents the OSCAR algorithm and its accelerated version Acc-OSCAR. Section V provides the performance guarantees for the proposed algorithm. Numerical simulations are given in Section VI, and conclusions are provided in Section VII.

Notations. For a natural number K , the symbol $[K]$ denotes $[K] = \{0, 1, \dots, K-1\}$. For a matrix $\mathbf{X} \in \mathbb{C}^{m \times n}$, let $\mathbf{X}^*$, $\mathbf{X}^T$, $\mathbf{X}^H$ denote the complex conjugate, transpose, and conjugate transpose, respectively. Let $\mathbf{I}_s$ denote the identity matrix with dimension $s \times s$. The symbol $\text{vec}(\mathbf{X})$ returns an $mn \times 1$ vector in column-major order, and $\text{diag}(\mathbf{A}) \in \mathbb{C}^{mn \times mn}$ is a diagonal matrix formed by arranging the elements of $\mathbf{A} \in \mathbb{C}^{m \times n}$ in column-major order. The inner product of two vectors is defined as $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^H \mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ and the inner product of two matrices is defined as $\langle \mathbf{X}, \mathbf{Y} \rangle = \text{tr}(\mathbf{X}^H \mathbf{Y}) = \text{vec}(\mathbf{X})^H \text{vec}(\mathbf{Y})$ for $\mathbf{X}, \mathbf{Y} \in \mathbb{C}^{m \times n}$. The largest entry of a matrix $\mathbf{X}$ is defined as $\|\mathbf{X}\|_\infty$. The ℓ_p norm of a vector $\mathbf{x}$ is termed as $\|\mathbf{x}\|_p$. The term $\|\cdot\|_{2,\infty}$ denotes the largest ℓ_2 -norm of its rows.

II. PROBLEM FORMULATION

This section formulates the optimization model by leveraging two properties of the estimated spectrally sparse signal. In particular, the model exploits the low-rankness of the lifting Hankel matrix of the estimated signal and the sign consistency between the original signal and the estimated signal as follows:

1) Low-rankness of lifting Hankel matrix. Previous work [18], [21] has shown that the lifting Hankel matrix $\mathcal{H}(\mathbf{x}^*) \in \mathbb{C}^{n_1 \times n_2}$ is of low rank, that is,

$$\text{rank}(\mathcal{H}(\mathbf{x}^*)) = r, \quad (5)$$

where $\mathcal{H}$ is the Hankel lifting operator that maps a vector $\mathbf{z} \in \mathbb{C}^n$ to a matrix $\mathcal{H}\mathbf{z} \in \mathbb{C}^{n_1 \times n_2}$ with $[\mathcal{H}\mathbf{z}]_{i,j} = z_{i+j}$, $i \in [n_1]$, $j \in [n_2]$ and $n_1 + n_2 - 1 = n$. Thus, we require that the estimated signal $\mathbf{z}$ satisfies

$$\text{rank}(\mathcal{H}(\mathbf{z})) = r. \quad (6)$$

2) Sign consistency. According to the definition of one-bit sampling, we have

$$\begin{pmatrix} \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{x}^*) \\ \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{x}^*) \end{pmatrix} \succeq 0, \quad (7)$$

which leads to

$$\text{Re}(\bar{\mathbf{y}} \odot \mathbf{x}^*) = \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{x}^*) + \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{x}^*) \succeq 0, \quad (8)$$

where $\odot$ denotes the Hadamard product, $\mathbf{a} \succeq 0$ means $a_k \geq 0, k = 1, \dots, n$, for $\mathbf{a} \in \mathbb{R}^n$. Therefore, to enforce the sign consistency of the one-bit sampling $\mathbf{y}$ and the estimated signal $\mathbf{z}$, the number of negative elements in $\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})$ should be as small as possible. Inspired by the binary iterative hard thresholding (BIHT) approach in standard compressed sensing [22], we use the convex relaxation called one-sided ℓ_p -norm to achieve this

$$\left\| [\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})]_- \right\|_p^p, \quad (9)$$

where $[\mathbf{q}]_-$ keeps the negative elements in $\mathbf{q} \in \mathbb{R}^n$ and sets the nonnegative elements in $\mathbf{q}$ to be zero, and $p \in \mathbb{Z}^+$ is a positive integer. Like [22] and [4], we choose $p = 1$ and $p = 2$ in our algorithm, where $p = 1$ is suitable for sparse noise and $p = 2$ is suitable for dense noise.

Remark 1. An alternative choice of the objective function (9) is

$$\left\| \begin{pmatrix} \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{z}) \\ \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{z}) \end{pmatrix} \right\|_p^p. \quad (10)$$

But it makes the algorithm and expressions more complicated and shows no benefit according to the simulations (c.f. Figs. 3 and 4), so we choose to use (9) to enforce the sign consistency.

Notice that Eqs. (5) and (8) hold for any scale of $\mathbf{x}^*$. To eliminate this scaling effect, we require its estimate $\mathbf{z}$ to satisfy

$$\|\mathcal{H}(\mathbf{z})\|_F^2 = 1 \quad (11)$$

for ease of analysis¹. Then the optimization problem becomes

$$\begin{aligned} \min_{\mathbf{z}} \quad & \left\| [\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})]_- \right\|_p^p \\ \text{s.t.} \quad & \text{rank}(\mathcal{H}(\mathbf{z})) = r, \|\mathcal{H}(\mathbf{z})\|_F^2 = 1. \end{aligned} \quad (12)$$

Given the complexity associated with the rank constraint (6), the problem presented is NP-hard. Given the fact that n

¹The right-hand side can be chosen as an arbitrary positive constant without influencing the estimation of frequencies and damping factors.

is odd, we employ symmetric Hankel matrix factorization as an alternative to Eq. (6), that is,

$$\mathcal{H}(z) = \mathbf{U}\mathbf{U}^T, \mathbf{U} \in \mathbb{C}^{s \times r}. \quad (13)$$

where $s = (n + 1)/2$ is an integer.

To enforce the Hankel structure of $\mathbf{U}\mathbf{U}^T$, we introduce some important operators. We denote $\mathcal{H}^*$ as the adjoint of $\mathcal{H}$, which maps a matrix $\mathbf{M} \in \mathbb{C}^{n_1 \times n_2}$ to a vector $\mathcal{H}^*(\mathbf{M}) = \{\sum_{i+j=a} \mathbf{M}_{i,j}\}_{a=0}^{n-1} \in \mathbb{C}^n$. Let $\mathcal{D}^2 = \mathcal{H}^* \mathcal{H}$ and $\mathcal{D}$ is a linear operator that maps a vector $\mathbf{q} \in \mathbb{C}^n$ to $\mathcal{D}\mathbf{q} \in \mathbb{C}^n$, where $[\mathcal{D}\mathbf{q}]_a = \sqrt{w_a} q_a$ and w_a is the length of the a -th skew-diagonal of an $n_1 \times n_2$ matrix with $a \in [n]$. Define $\mathcal{G} = \mathcal{H}\mathcal{D}^{-1}$. Since $\mathcal{D}$ is invertible, we have $\mathcal{G}^* \mathcal{G} = \mathcal{I}$, where $\mathcal{I}$ denotes the identity operator. Then we can enforce the Hankel structure [3] as follows

$$(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) = 0, \quad (14)$$

where $\mathcal{I} - \mathcal{G}\mathcal{G}^*$ is a projection operator.

Since $\mathcal{G}^* \mathcal{G} = \mathcal{I}$ and $\mathcal{H} = \mathcal{G}\mathcal{D}$, we have $\mathbf{z} = \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)$. Therefore, we reformulate the optimization problem as

$$\begin{aligned} \min_{\mathbf{U}} \quad & \left\| \text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)] \right\|_p^p \\ \text{s.t.} \quad & (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T) = 0, \|\mathbf{U}\mathbf{U}^T\|_F^2 = 1. \end{aligned} \quad (15)$$

We aim to find the best rank- r approximation that minimizes the negative sign error and satisfies the Hankel structure and the norm constraint. Following the previous work [23], [3], we constrain the solution $\mathbf{U}$ in a convex set $\mathcal{C}$, which is defined as

$$\mathcal{C} = \left\{ \mathbf{U} \in \mathbb{C}^{s \times r} \mid \|\mathbf{U}\|_{2,\infty} \leq 2\sqrt{\frac{\mu r \sigma}{n}} \right\}, \quad (16)$$

where $\mu > 0$ is an incoherence constant and $\sigma = \sigma_1(\mathcal{H}(\mathbf{x}^*))$ denotes the maximum singular value of $\mathcal{H}(\mathbf{x}^*)$.

Finally, we penalize the first constraint in the objective function, which is formulated as

$$\begin{aligned} \min_{\mathbf{U}} \quad & F_p(\mathbf{U}) \triangleq \frac{1}{p} \left\| \text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)] \right\|_p^p \\ & + \gamma \|(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\|_F^2 \\ \text{s.t.} \quad & \mathbf{U} \in \mathcal{C}, \|\mathbf{U}\mathbf{U}^T\|_F^2 = 1. \end{aligned} \quad (17)$$

where $\gamma > 0$ is a tradeoff parameter. Define

$$f_p(\mathbf{U}) = \frac{1}{p} \left\| \text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)] \right\|_p^p, \quad (18)$$

$$g(\mathbf{U}) = \|(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\|_F^2. \quad (19)$$

Then we have

$$F_p(\mathbf{U}) = f_p(\mathbf{U}) + \gamma g(\mathbf{U}). \quad (20)$$

It should be noted that $[\cdot]_-$ is non-smooth, and both the objective function and the constraint are nonconvex. Therefore, Problem (17) is a nonsmooth and nonconvex problem. The absence of smoothness impedes the application of momentum acceleration techniques for algorithm design and makes it difficult to analyze the convergence of the corresponding algorithms. To address this issue, a smooth approximation of the objective function is introduced in Section III.

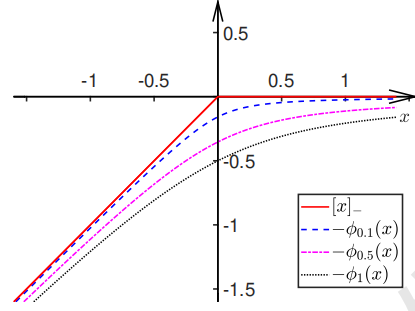


Fig. 2. Comparisons of $[x]_-$ and its smoothed function $-\phi_b(x)$ under different smoothing parameters b .

III. SMOOTH APPROXIMATION

In this section, we study the smooth approximation of Problem (17) to make it easy for the acceleration algorithm design and convergence analysis.

Note that $[q]_- = -\max(-q, 0) = -\text{ReLU}(-q)$, where ReLU is a classical activation function in deep learning. Motivated by the smoothing technique of ReLU [24], we propose to use a smooth function $-\phi_b(q)$ to replace the original one-sided negative function $[q]_-$, which is an element-wise function defined as

$$[\phi_b(\mathbf{q})]_i = \phi_b(q_i) = \frac{\sqrt{q_i^2 + b} - q_i}{2}, \quad (21)$$

where $b > 0$ is a smoothing parameter. When $b = 0$, $-\phi_b(q)$ degrades to $[q]_-$. As shown in Fig. 2, $-\phi_b(q)$ approximates $[q]_-$ from below, and the smaller b is, the closer $-\phi_b(q)$ is to $[q]_-$.

By simple calculation, for any $q_i \in \mathbb{R}$, we obtain

$$\phi_b(q_i) > 0, \quad (22)$$

$$\phi'_b(q_i) = \frac{1}{2} \left(\frac{q_i}{\sqrt{q_i^2 + b}} - 1 \right) < 0, \quad (23)$$

$$\phi''_b(q_i) = \frac{b}{2(q_i^2 + b)^{\frac{3}{2}}} > 0, \quad (24)$$

where $\phi'_b(q)$ and $\phi''_b(q)$ denotes the first-order gradient and second-order gradient of ϕ_b at $q \in \mathbb{R}$, respectively.

Therefore, we can change the optimization problem from the nonsmooth counterpart to the following smooth one

$$\begin{aligned} \min_{\mathbf{U}} \quad & \hat{F}_p(\mathbf{U}) \triangleq \frac{1}{p} \left\| \phi_b(\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]) \right\|_p^p \\ & + \gamma \|(\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\|_F^2 \\ \text{s.t.} \quad & \mathbf{U} \in \mathcal{C}, \|\mathbf{U}\mathbf{U}^T\|_F^2 = 1. \end{aligned} \quad (25)$$

Define

$$\hat{f}_p(\mathbf{U}) = \frac{1}{p} \left\| \phi_b(\text{Re}[\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)]) \right\|_p^p, \quad (26)$$

then we have

$$\hat{F}_p(\mathbf{U}) = \hat{f}_p(\mathbf{U}) + \gamma g(\mathbf{U}). \quad (27)$$

Compared with Problem (17), there are two main benefits for Problem (25): 1) The objective function of (25) is smooth, we can adopt momentum-based approaches to accelerate the algorithm if necessary; 2) While the convergence analysis is

difficult for the nonconvex and nonsmooth objective function $F_p(\mathbf{U})$, it is relatively simple for the nonconvex but smooth objective function $\hat{F}_p(\mathbf{U})$.

In the subsequent sections, we will propose the algorithms for Problem (25) in Section IV and give the related convergence analysis in Section V.

IV. ALGORITHM

This section proposes a one-bit spectral compressed sensing algorithm via Hankel matrix factorization (OSCAR) for problem (25) and its accelerated variant Acc-OSCAR. First of all, we present the initialization scheme. The initial point $\mathbf{U}^0$ is set as follows: Let the Takagi factorization of $\mathcal{H}(\mathbf{y})$ be

$$\mathcal{T}_r(\mathcal{H}(\mathbf{y})) = \mathbf{U}_y \mathbf{S}_y \mathbf{U}_y^T, \quad (28)$$

where $\mathbf{U}_y \in \mathbb{C}^{s \times r}$ and $\mathbf{S}_y \in \mathbb{C}^{r \times r}$. Then we set

$$\mathbf{U}^0 = \mathcal{P}_C(\mathbf{U}_y \mathbf{S}_y^{\frac{1}{2}}), \quad (29)$$

where the projection is defined as

$$\mathcal{P}_C(\mathbf{U}) = \arg \min_{\tilde{\mathbf{U}} \in C} \|\mathbf{U} - \tilde{\mathbf{U}}\|_F, \quad (30)$$

that is,

$$[\mathcal{P}_C(\mathbf{U})]_{i,:} = \begin{cases} \mathbf{U}_{i,:}, & \text{if } \|\mathbf{U}_{i,:}\|_2 \leq 2\sqrt{\frac{\mu r \sigma}{n}}, \\ \frac{\mathbf{U}_{i,:}}{\|\mathbf{U}_{i,:}\|_2} 2\sqrt{\frac{\mu r \sigma}{n}}, & \text{otherwise,} \end{cases} \quad (31)$$

and $\mathbf{U}_{i,:}$ is the i -th row of $\mathbf{U}$.

Then we use gradient-based methods to update the iteration variable. Before providing the gradients, we review some important facts from [25].

Fact 1 (Proposition 3.4, [25]). *For a scalar function $h(\mathbf{S}, \bar{\mathbf{S}}) : \mathbb{C}^{m \times n} \times \mathbb{C}^{m \times n} \rightarrow \mathbb{C}$, if the matrix differential of the function is*

$$dh(\mathbf{S}, \bar{\mathbf{S}}) = \langle \mathbf{A}, d\mathbf{S} \rangle + \langle \mathbf{B}, d\bar{\mathbf{S}} \rangle, \quad (32)$$

then its complex gradient matrices are the conjugate of the matrices $\mathbf{A}$ and $\mathbf{B}$, i.e.,

$$\nabla_{\mathbf{S}} h(\mathbf{S}, \bar{\mathbf{S}}) = \bar{\mathbf{A}}, \text{ and } \nabla_{\bar{\mathbf{S}}} h(\mathbf{S}, \bar{\mathbf{S}}) = \bar{\mathbf{B}}. \quad (33)$$

Fact 2 (Theorem 4.4, [25]). *For a real function $h(\mathbf{S}, \bar{\mathbf{S}}) : \mathbb{C}^{m \times n} \times \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$, its fastest growing direction is given by $\nabla_{\bar{\mathbf{S}}} h(\mathbf{S}, \bar{\mathbf{S}})$.*

Based on Fact 2, the gradients of $\hat{F}_p(\mathbf{U})$, $\hat{f}_p(\mathbf{U})$ and $g(\mathbf{U})$ under Wirtinger calculus are

$$\nabla F_p(\mathbf{U}) \triangleq \nabla \hat{F}_p(\mathbf{U}) + \gamma \nabla g(\mathbf{U}), \quad (34)$$

$$\nabla \hat{f}_p(\mathbf{U}) \triangleq \nabla_{\bar{\mathbf{U}}} \hat{f}_p(\mathbf{U}), \quad (35)$$

$$\nabla g(\mathbf{U}) \triangleq \nabla_{\bar{\mathbf{U}}} g(\mathbf{U}). \quad (36)$$

It is easy to get the gradient of $g(\mathbf{U})$, which is

$$\nabla g(\mathbf{U}) = \nabla_{\bar{\mathbf{U}}} g(\mathbf{U}) = (\mathbf{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\bar{\mathbf{U}}. \quad (37)$$

Next, we provide the gradients of $\hat{f}_p(\mathbf{U})$ under Wirtinger calculus for $p = 1, 2$.

Lemma 1. *Define $\mathbf{w} = \text{Re}(\bar{\mathbf{y}} \odot \mathcal{D}^{-1}\mathcal{G}^*(\mathbf{U}\mathbf{U}^T))$. Then*

(a) *For $p = 1$, the gradient of $\hat{f}_p(\mathbf{U})$ is*

$$\nabla \hat{f}_1(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}. \quad (38)$$

(b) *For $p = 2$, the gradient of $\hat{f}_p(\mathbf{U})$ is*

$$\nabla \hat{f}_2(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}. \quad (39)$$

Proof. The proof is provided in Appendix E. $\square$

According to Lemma 1 and Eq. (37), we obtain

$$\nabla \hat{F}_1(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}} + \gamma(\mathbf{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\bar{\mathbf{U}}, \quad (40)$$

$$\nabla \hat{F}_2(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}} + \gamma(\mathbf{I} - \mathcal{G}\mathcal{G}^*)(\mathbf{U}\mathbf{U}^T)\bar{\mathbf{U}}. \quad (41)$$

Now, we are ready to describe our algorithm flow. The algorithm is provided in Algorithm 1. The maximum number of rounds is K . In each round k , the update rule is

$$\mathbf{U}^{k+1} = \mathcal{P}_C(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k)), \quad (42)$$

where $\eta > 0$ is the step size. In the last round, we map the lifting matrix back to the signal

$$\mathbf{z}^K = \mathcal{D}^{-1}\mathcal{G}^*\left(\frac{\mathbf{U}^K(\mathbf{U}^K)^T}{\|\mathbf{U}^K(\mathbf{U}^K)^T\|_F}\right). \quad (43)$$

Empirical results and previous works [22], [26] demonstrate that normalizing the signal at the last round is enough for one-bit problems.

Based on Algorithm 1, we apply the momentum acceleration strategy to design the accelerated version in Algorithm 2.

$$\mathbf{U}^{k+1} = \mathcal{P}_{C'}(\tilde{\mathbf{U}}^k - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k)), \quad (44)$$

$$\tilde{\mathbf{U}}^{k+1} = \mathbf{U}^{k+1} + \beta(\mathbf{U}^{k+1} - \mathbf{U}^k), \quad (45)$$

where $\tilde{\mathbf{U}} \in \mathbb{C}^{s \times r}$ denotes the extrapolated matrix, $\beta > 0$ is an extrapolated parameter and the set C' is defined as

$$C' = \left\{ \mathbf{U} \in \mathbb{C}^{s \times r} \mid \|\mathbf{U}\|_{2,\infty} \leq \frac{2}{1+2\beta} \sqrt{\frac{\mu r \sigma}{n}} \right\}. \quad (46)$$

Here we introduce a smaller set C' to ensure that the iterates $\mathbf{U}^k, \tilde{\mathbf{U}}^k$ fall into the incoherence set C .

Besides, we modify the traditional accelerated method to ensure a sufficient decrease in the objective function, thereby achieving more stable convergence behavior [27], [28]. As shown in Algorithm 2, the update rule enforces a monotonic decrease by accepting the new iterate only when the objective function value is reduced.

Fast computation. We present the fast computation of the proposed algorithms. According to [18], [3], two kinds of terms can be computed quickly:

- 1) For any intermediate vector $\mathbf{q} \in \mathbb{C}^n$, the term $\mathcal{G}(\mathbf{q})\bar{\mathbf{U}}$ can be computed by fast Hankel matrix-vector multiplication, which is realized by r fast convolutions with $O(nr \log(n))$ flops via FFT.
- 2) The term $\mathcal{G}^*(\mathbf{U}\mathbf{U}^T)$ can be computed quickly by r fast convolution with $O(nr \log(n))$ flops via FFT.

The computational complexity for (37) is also $O(nr \log(n))$ [18], [3]. Additionally, the projection operation can be integrated into the computation involved in fast convolution,

Algorithm 1 OSCAR- ℓ_p : One-Bit Spectral Compressed Sensing via Hankel Matrix Factorization

Input: p, γ, η
Initialization: $\mathcal{T}_r(\mathcal{H}(y)) = U_y S_y U_y^H, U^0 = \mathcal{P}_C(U_y S_y^{\frac{1}{2}})$
for $k = 0, 1, \dots, K-1$ **do**
 $U^{k+1} = \mathcal{P}_C(U^k - \eta \nabla \hat{F}_p(U^k))$
end for
Output: $U^K, z^K = \mathcal{D}^{-1} \mathcal{G}^*(U^K (U^K)^T / \|U^K (U^K)^T\|_F)$

Algorithm 2 Acc-OSCAR- ℓ_p : Accelerated One-Bit Spectral Compressed Sensing via Hankel Matrix Factorization

Input: p, γ, η, β
Initialization: $\mathcal{T}_r(\mathcal{G}(y)) = U_y S_y U_y^H, U^0 = \tilde{U}^0 = \mathcal{P}_{C'}(U_y S_y^{\frac{1}{2}})$
for $k = 0, 1, \dots, K-1$ **do**
 $U^{k+1} = \mathcal{P}_{C'}(\tilde{U}^k - \eta \nabla \hat{F}_p(\tilde{U}^k))$
 $V^{k+1} = U^{k+1} + \beta(U^{k+1} - U^k)$
if $\hat{F}_p(U^{k+1}) \leq \hat{F}_p(V^{k+1})$ **then**
 $\tilde{U}^{k+1} = U^{k+1}$
else if $\hat{F}_p(V^{k+1}) < \hat{F}_p(U^{k+1})$ **then**
 $\tilde{U}^{k+1} = V^{k+1}$
end for
Output: $U^K, z^K = \mathcal{D}^{-1} \mathcal{G}^*(U^K (U^K)^T / \|U^K (U^K)^T\|_F)$

thus eliminating the need to separately account for the computational complexity of projecting onto $\mathcal{C}$. As a result, the computational complexity of each iteration of our algorithms is $O(nr \log(n))$.

V. CONVERGENCE ANALYSIS

This section gives the convergence analysis for Algorithm 1 and Algorithm 2. Before giving the theoretical analysis, we first introduce some useful facts and lemmas. Subsequently, we will provide the Hessian matrices and prove the smoothness of the objective functions. Finally, we analyze the convergence property of the two algorithms.

Define the orthonormal bases for $s \times s$ Hankel matrix

$$H_a = \frac{1}{\sqrt{w_a}} \mathcal{H}(e_a), a = 1, \dots, n. \quad (47)$$

For any Hankel matrix $\mathcal{H}(z)$, we have

$$\mathcal{H}(z) = \sum_{a=1}^n \langle H_a, \mathcal{H}(z) \rangle H_a. \quad (48)$$

In addition, $\mathcal{G}\mathcal{G}^*(W)$ can be reformulated as

$$\begin{aligned} \mathcal{G}\mathcal{G}^*(W) &= \sum_{a=1}^n \langle H_a, \mathcal{G}\mathcal{G}^*(W) \rangle H_a \\ &= \sum_{a=1}^n \langle \mathcal{G}\mathcal{G}^*(H_a), W \rangle H_a \\ &= \sum_{a=1}^n \langle H_a, W \rangle H_a. \end{aligned} \quad (49)$$

Next, we review one important fact about the relationship between the second-order matrix differential and the Hessian matrix, which is reformulated for our proof.

Fact 3 (Theorem 3.4, [25]). For a differentiable real function $h(S, \bar{S}) : \mathbb{C}^{m \times n} \times \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$, if its second-order matrix differential is

$$\begin{aligned} d^2 h(S, \bar{S}) &= (\text{dvec}(S))^T C \text{dvec}(S) \\ &\quad + (\text{dvec}(\bar{S}))^T D \text{dvec}(S) \\ &\quad + (\text{dvec}(S))^T \bar{D} \text{dvec}(\bar{S}) \\ &\quad + (\text{dvec}(\bar{S}))^T \bar{C} \text{dvec}(\bar{S}), \end{aligned} \quad (50)$$

then the full Hessian matrix of $h(S, \bar{S})$ is

$$\nabla^2 h(S, \bar{S}) = \begin{bmatrix} H_{\bar{S}, S} & H_{\bar{S}, \bar{S}} \\ H_{S, S} & H_{S, \bar{S}} \end{bmatrix}, \quad (51)$$

where

$$H_{S, S} = \frac{1}{2}(C + C^T), \quad H_{S, \bar{S}} = \frac{1}{2}(D + D^H), \quad (52)$$

$$H_{\bar{S}, \bar{S}} = \overline{H_{S, S}}, \quad H_{\bar{S}, S} = \overline{H_{S, \bar{S}}}. \quad (53)$$

Then we provide the Hessian matrices for $\hat{F}_1(U)$ and $\hat{F}_2(U)$, respectively.

Lemma 2. Define $w = \text{Re}(\bar{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(UU^T))$. Then we have:

(a) For $p = 1$, the Hessian matrix of $\hat{F}_p(U)$ is

$$\nabla^2 \hat{F}_1(U) = \begin{bmatrix} H_{\bar{U}, U}^{\hat{F}_1} & H_{\bar{U}, \bar{U}}^{\hat{F}_1} \\ H_{U, U}^{\hat{F}_1} & H_{U, \bar{U}}^{\hat{F}_1} \end{bmatrix} \quad (54)$$

with

$$\begin{aligned} H_{\bar{U}, U}^{\hat{F}_1} &= (U^T \otimes I_s) \Gamma_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi_b''(w))) (\bar{U} \otimes I_s) \\ &\quad + \gamma(U^T \otimes I_s) \Gamma_g (\bar{U} \otimes I_s), \end{aligned} \quad (55)$$

$$\begin{aligned} H_{\bar{U}, \bar{U}}^{\hat{F}_1} &= I_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi_b'(w) \odot y) + \frac{1}{2}(U^H \otimes I_s) \Gamma_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi_b''(w) \odot y \odot y)) (\bar{U} \otimes I_s) \\ &\quad + \gamma I_r \otimes (I - \mathcal{G}\mathcal{G}^*)(UU^T), \end{aligned} \quad (56)$$

$$H_{U, \bar{U}}^{\hat{F}_1} = \overline{H_{\bar{U}, U}^{\hat{F}_1}}, \quad H_{U, U}^{\hat{F}_1} = \overline{H_{\bar{U}, \bar{U}}^{\hat{F}_1}}, \quad (57)$$

where

$$\Gamma_g = (I_{s^2} + K_{s^2}) \left(I_{s^2} - \sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right), \quad (58)$$

$$\Gamma_f = (I_{s^2} + K_{s^2}) \left(\sum_{a=1}^n \text{vec}(H_a) \text{vec}(H_a)^T \right), \quad (59)$$

and $K_{s^2} \in \mathbb{R}^{s^2 \times s^2}$ denotes the perfect shuffle matrix that can convert column-major vectorization to row-major vectorization [29].

(b) For $p = 2$, the Hessian matrix of $\hat{F}_p(U)$ is

$$\nabla^2 \hat{F}_2(U) = \begin{bmatrix} H_{\bar{U}, U}^{\hat{F}_2} & H_{\bar{U}, \bar{U}}^{\hat{F}_2} \\ H_{U, U}^{\hat{F}_2} & H_{U, \bar{U}}^{\hat{F}_2} \end{bmatrix} \quad (60)$$

with

$$\begin{aligned} H_{\bar{U}, U}^{\hat{F}_2} &= (U^T \otimes I_s) \Gamma_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(w))) (\bar{U} \otimes I_s) \\ &\quad + \gamma(U^T \otimes I_s) \Gamma_g (\bar{U} \otimes I_s), \end{aligned} \quad (61)$$

$$\begin{aligned} H_{\bar{U}, \bar{U}}^{\hat{F}_2} &= I_r \otimes \mathcal{G} \mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2}(\mathbf{U}^H \otimes \mathbf{I}_s) \\ &\quad \cdot \Gamma_f \text{diag} \left(\mathcal{G} \mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y}) \right) (\bar{\mathbf{U}} \otimes \mathbf{I}_s) \\ &\quad + \gamma \mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G} \mathcal{G}^*)(\mathbf{U} \mathbf{U}^T), \end{aligned} \quad (62)$$

$$H_{\bar{U}, \bar{U}}^{\hat{F}_2} = \overline{H_{\bar{U}, \bar{U}}^{\hat{F}_2}}, \quad H_{\bar{U}, \bar{U}}^{\hat{F}_2} = \overline{H_{\bar{U}, \bar{U}}^{\hat{F}_2}}, \quad (63)$$

where

$$\Lambda_\phi(\mathbf{w}) = \phi'_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) + \phi_b(\mathbf{w}) \odot \phi''_b(\mathbf{w}). \quad (64)$$

Proof. The proof is provided in Appendix H of the Supplementary Material. $\square$

Next, we present that the objective function is smooth by proving that the Hessian Matrices are upper bounded.

Lemma 3. For any $\mathbf{U} \in \mathcal{C}$, where $\mathcal{C}$ is the set defined in (16), the spectral norm of $\nabla^2 \hat{F}_p(\mathbf{U})$ are upper-bounded, that is,

$$\|\nabla^2 \hat{F}_p(\mathbf{U})\| \leq L_p, \quad (65)$$

where L_p is defined as

$$L_p = \begin{cases} 2\sqrt{2n} + 8\mu r \sigma(\frac{2}{\sqrt{b}} + 3\gamma), & p = 1, \\ 8(\sqrt{\mu^2 r^2 \sigma^2 / n + n b} + \mu r \sigma / \sqrt{n}) + \mu r \sigma(9 + 3\gamma), & p = 2. \end{cases}$$

Proof. The proof is provided in Appendix F. $\square$

Finally, we theoretically establish the convergence of the algorithms toward an approximate first-order stationary point. Based on the update rule in Eqs. (42) and (44), the first-order optimality condition [30], [31] means that there exists $\mathbf{U} \in \mathbb{C}^{n \times r}$ such that

$$-\nabla \hat{F}_p(\mathbf{U}) \in N_{\mathcal{C}}(\mathbf{U}), \quad (66)$$

and the approximate first-order optimality condition means that there exists $\mathbf{U} \in \mathbb{C}^{n \times r}$ such that

$$-\nabla \hat{F}_p(\mathbf{U}) \in N_{\mathcal{C}}(\mathbf{U}) + \mathcal{B}_2(0, \varepsilon), \quad (67)$$

where $N_{\mathcal{C}}(\mathbf{U})$ denotes the normal cone of $\mathcal{C}$ at $\mathbf{U}$ and $\mathcal{B}_2(\mathbf{z}, r) \triangleq \{\mathbf{q} \in \mathbb{C}^d : \|\mathbf{q} - \mathbf{z}\|_2 \leq r\}$ denotes the Euclidean ball of radius r centered at $\mathbf{z}$.

Theorem 1. For the iterates in Algorithm 1 the limit of the loss function satisfies

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \hat{F}_p^*, \quad (68)$$

and the sequence of iterations satisfies

$$\min_{k \in [K]} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F \leq \sqrt{\frac{2\hat{F}_p(\mathbf{U}^0)}{L_p K}}, \quad (69)$$

as long as $0 < \eta \leq \frac{1}{L_p}$, where $\hat{F}_p^*$ denotes some positive constant.

Proof. The proof is presented in Appendix A. $\square$

Theorem 2. Assume that K is large enough satisfying

$$K \geq \frac{8L_p \hat{F}_p(\mathbf{U}^0)}{\varepsilon^2}, \quad (70)$$

and $\eta = \frac{1}{L_p}$, then there exists $k' \in [K]$ in Algorithm 1 such that it satisfies the approximate first-order optimality condition

$$-\nabla \hat{F}_p(\mathbf{U}^{k'+1}) \in N_{\mathcal{C}}(\mathbf{U}^{k'+1}) + \mathcal{B}_2(0, \varepsilon). \quad (71)$$

Proof. The proof is shown in Appendix B. $\square$

Remark 2. Theorem 1 establishes that the sequence of iterations converges to a certain limit, with the minimum of the difference $\mathbf{U}^{k+1} - \mathbf{U}^k$ having an upper bound. Furthermore, Theorem 2 determines the iteration complexity required to satisfy the approximate first-order optimality condition, which is inversely proportional to the square of the error ε . Moreover, the iteration complexity is directly proportional to the initial function value $\hat{F}_p(\mathbf{U}^0)$. Therefore, effective initialization contributes to faster convergence to the approximate first-order optimality condition.

Theorem 3. Assume $\eta = \frac{1}{L_p}$, the sequence $\{\mathbf{U}^k\}$ generated by Algorithm 2 satisfies

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \hat{F}_p^\circ. \quad (72)$$

Let $\lim_{k \rightarrow \infty} \{\mathbf{U}^k\} = \mathbf{U}' \in \Omega$, then the set of limit points Ω is compact, and $\mathbf{U}'$ satisfies the first-order optimality condition that

$$-\nabla \hat{F}_p(\mathbf{U}') \in N_{\mathcal{C}}(\mathbf{U}').$$

Proof. See Appendix C. $\square$

Definition 1 (Distance). The distance of a point $\mathbf{x} \in \mathbb{C}^d$ to a closed set $\Theta \subseteq \mathbb{C}^d$ is defined as

$$\text{dist}_\Theta(\mathbf{x}) = \min_{\mathbf{y} \in \Theta} \|\mathbf{y} - \mathbf{x}\|_2.$$

We introduce the Kurdyka-Lojasiewicz (KL) property [32], [27], which is important to derive the convergence rate of the accelerated algorithm (Algorithm 2).

Assumption 1. The function h is said to satisfy the uniformized Kurdyka-Lojasiewicz (KL) property [32], [27] if

$$\varphi'(h(\mathbf{x}) - h^*) \text{dist}_{\partial h(\mathbf{x})}(\mathbf{0}) \geq \frac{1}{c_{\text{KL}}},$$

where c_{KL} is the KL constant, $\varphi(t) = 2t^{\frac{1}{2}}$, and h^* is some limit value of the function h .

Finally, we introduce an auxiliary function that plays an important role in establishing the convergence rate property of the Algorithm 2

$$G(\mathbf{U}) = \hat{F}_p(\mathbf{U}) + I_{\mathcal{C}'}(\mathbf{U}),$$

where $\mathbf{U} \in \mathbb{C}^{n \times r}$, $I_{\mathcal{C}'}$ is an indicator function such that $I_{\mathcal{C}'}(\mathbf{U}) = \infty$ for $\mathbf{U} \notin \mathcal{C}'$ and $I_{\mathcal{C}'}(\mathbf{U}) = 0$ for $\mathbf{U} \in \mathcal{C}'$.

Theorem 4. Suppose $G(\mathbf{U}^k)$ satisfies the KL property in Assumption 1 for $k \in [K]$:

$$\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\circ) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0}) \geq \frac{1}{c_{\text{KL}}}, \quad (73)$$

where $\hat{F}_p^\circ$ is the limit value defined in (72). For $\eta = \frac{1}{L_p}$, the sequence of iterations in Algorithm 2 satisfies

$$\hat{F}_p(\mathbf{U}^k) - \hat{F}_p^\circ \leq \left(1 - \frac{1}{1 + C_p}\right)^k (\hat{F}_p(\mathbf{U}^0) - \hat{F}_p^\circ),$$

where $C_p = 8c_{\text{KL}}^2 L_p$.

Proof. See Appendix D. $\square$

Remark 3. Theorem 3 establishes that the sequence of iterations in Algorithm 2 converges to a certain limit, and the set of limit points satisfies the first-order optimality condition. Theorem 4 gives the linear convergence rate of Algorithm 2 under the KL assumption. It is generally difficult to analyze the KL property of an arbitrary function, since it is often unclear whether the function is semi-algebraic [28], [33]. Moreover, the analysis of the KL property relies on the explicit expression of $\text{dist}_{\partial h(\mathbf{x})}(\mathbf{0})$ in Assumption 1 which is challenging to derive in practice. Therefore, we instead verify the stable KL property of Algorithm 2 empirically through simulations, as reported in Appendix G of the Supplementary Material.

VI. SIMULATIONS

This section provides the numerical simulations on one-bit DoA estimation with one snapshot for the proposed OSCAR algorithms and then compares two best of them with previous advanced algorithms, including CBIHT- ℓ_1 [4], CBIHT- ℓ_2 [4], iCBIHT [5], and 1bit-ANM [34].

DoA estimation studies the problem of estimating the angle of targets by the received signals of multiple sensors. Considering the DoA estimation of a uniform linear array of n sensors

$$\tilde{x}_t = \sum_{a=1}^r d_a \exp\left(-i2\pi t \sin(\theta_a) \frac{d}{\lambda}\right), t = 1, 2, \dots, n, \quad (74)$$

where $\theta_a \in [-90^\circ, 90^\circ]$ denotes the direction of arrivals and $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_r]$, d denotes the spacing between two adjacent sensors, and λ denotes the wavelength of the signals. Notice that spectral compressed sensing becomes DoA estimation by setting $f_a = \frac{(\sin(\theta_a)+1)d}{\lambda}$, $\tau_a = \frac{d}{\lambda}$, and $d = \frac{\lambda}{2}$ in Eq. (11). The performance is assessed under Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{r} \sum_{a=1}^r \|\hat{\theta}_a - \theta_a\|_2^2}, \quad (75)$$

where $\hat{\theta}_a$ denotes the estimation of θ_a .

A. Comparisons for the proposed OSCAR algorithms

In this section, we first compare the performance for OSCAR- ℓ_1 , OSCAR- ℓ_2 , Acc-OSCAR- ℓ_1 , Acc-OSCAR- ℓ_2 , Acc-COSCAR- ℓ_1 , and Acc-COSCAR- ℓ_2 in Figs. 3 and 4. Here, Acc-COSCAR- ℓ_1 and Acc-COSCAR- ℓ_2 correspond to the accelerated algorithms Acc-OSCAR- ℓ_1 , Acc-OSCAR- ℓ_2 by replacing $\text{Re}(\bar{\mathbf{y}} \odot \mathbf{z})$ with

$$\begin{pmatrix} \text{Re}(\mathbf{y}) \odot \text{Re}(\mathbf{z}) \\ \text{Im}(\mathbf{y}) \odot \text{Im}(\mathbf{z}) \end{pmatrix}. \quad (76)$$

For brevity, we omit the related details.

The dimension of $\mathbf{x}^*$ is set to be $n = 63$, the number of sinusoids is $r = 2$, $\boldsymbol{\theta} = [20.3^\circ, 22.5^\circ]$, the tradeoff parameter is $\gamma = 100$, the step size is $\eta = 0.001$, the smoothness parameter is $b = 0.001$, the momentum parameter is $\beta = 0.7$,

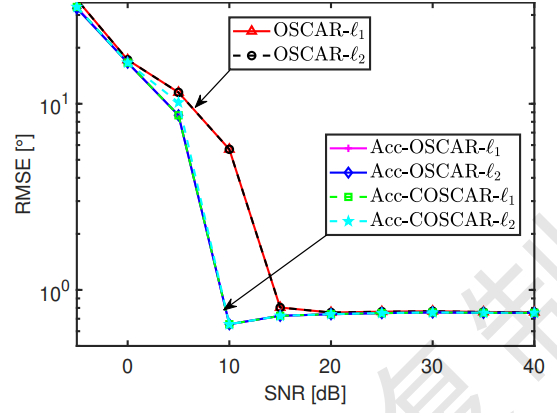


Fig. 3. RMSE v.s. SNR for all OSCARs.

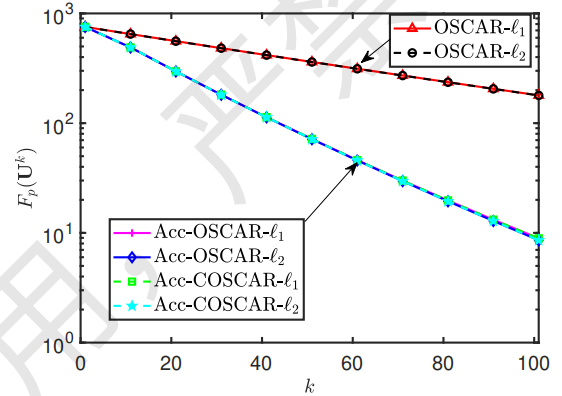


Fig. 4. Objective function $F_p(\mathbf{U}^k)$ v.s. k for all OSCARs.

and the maximum number of iterations is $K = 100$. One hundred Monte Carlo trials were conducted to get the RMSE.

Fig. 3 shows the RMSE as a function of SNR and Fig. 4 presents the objective function $F_p(\mathbf{U}^k)$ as a function of k for all OSCAR algorithms for SNR = 10 dB. The results indicate that the performance of all accelerated algorithms is nearly identical, and they exceed the performance of non-accelerated algorithms. This observation corroborates the statement in Remark 1 and confirms the efficacy of the acceleration technique.

B. DOA estimation with one snapshot

Since Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 outperform other OSCAR algorithms, we compare our Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 with CBIHT, iCBIHT, and 1bit-ANM in this subsection. The parameters for all other algorithms are set as recommended.

First of all, we compare the success rate and the RMSE for all successful trials of all algorithms with $n = 63$, $r = 2$ and $\boldsymbol{\theta} = [-20.3^\circ, 24.5^\circ]$, and the maximum number of iterations is $K = 100$. A trial is called successful if its RMSE satisfies

$$\text{RMSE} < 3^\circ. \quad (77)$$

One hundred Monte Carlo trials were conducted to get the success rate and RMSE. The rationale behind introducing the success rate is that there is a possibility that the DoAs cannot be recovered in certain trials under a one-bit single-measurement vector and low SNR. We carefully choose a

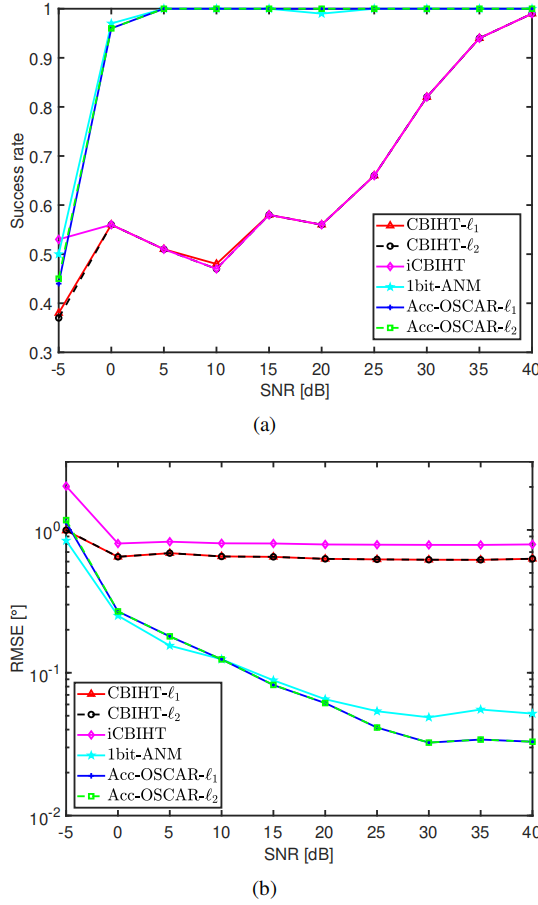


Fig. 5. Comparisons for different algorithms with $n = 63$ and $r = 2$. (a) Success Rate v.s. SNR. (b) RMSE v.s. SNR.

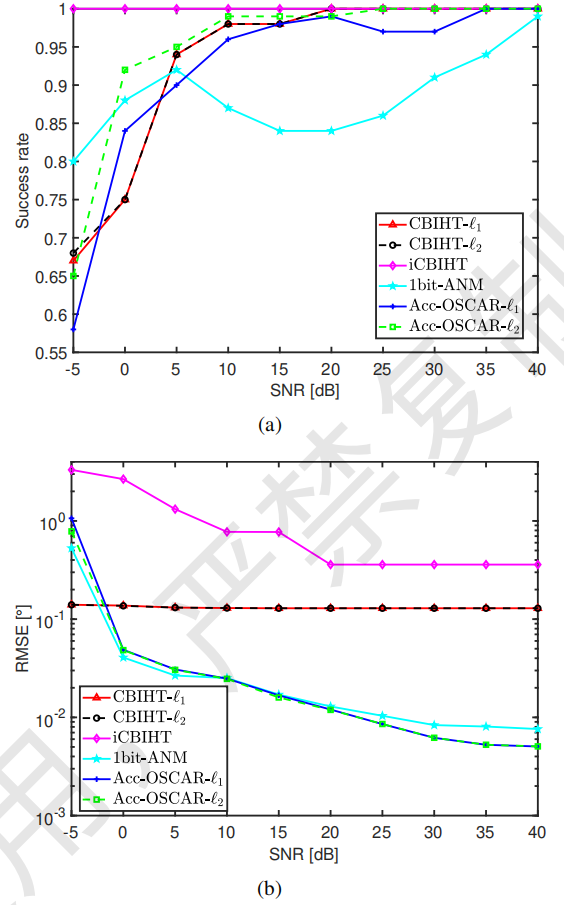


Fig. 6. Comparisons for different algorithms with $n = 255$ and $r = 3$. (a) Success Rate v.s. SNR. (b) RMSE v.s. SNR.

relatively small number 3° to facilitate easy comparison in the plot.

Fig. 5(a) shows that our algorithms Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 have similar success rates with 1-bit ANM, which performs much better than CBIHT- ℓ_1 , CBIHT- ℓ_2 and iCBIHT. Fig. 5(b) shows that Acc-OSCAR- ℓ_1 and Acc-OSCAR- ℓ_2 have smaller RMSE than other algorithms, especially when the SNR is larger than 20 dB.

Then we conduct simulations for a large number of sensors, where $n = 255$, $r = 3$ and $\theta = [-56.3^\circ, -15.4^\circ, 30.5^\circ]$, and the maximum number of iterations is $K = 100$. As illustrated in Fig. 6, Acc-OSCAR- ℓ_2 demonstrates superior performance compared to other advanced algorithms, since Acc-OSCAR- ℓ_2 has a better success rate curve except for the iCBIHT algorithm, and nearly achieves the best RMSE. The enhanced performance of Acc-OSCAR- ℓ_2 over Acc-OSCAR- ℓ_1 can be attributed to the nature of the noise, which is dense rather than sparse. Additionally, although iCBIHT has a 100% success rate, it suffers from poor RMSE since iCBIHT is designed for multiple snapshots, whereas this particular setting involves only a single snapshot.

VII. CONCLUSIONS

This paper proposes a gridless one-bit single-measurement spectral compressed sensing algorithm based on low-rank

Hankel matrix factorization. We establish a non-convex optimization model using the low-rank property of the Hankel matrix and one-sided ℓ_p norm. A gradient descent algorithm, OSCAR, are designed with the smoothing technique, and its accelerated version, Acc-OSCAR, is designed by using momentum techniques. Theoretical analysis and experiments validate its convergence in single-measurement scenarios, confirming the algorithm's effectiveness.

APPENDIX A PROOF OF THEOREM II

Note that $\mathbf{U}^{k+1}$ is the solution to the following optimization problem.

$$\mathbf{U}^{k+1} = \arg \min_{\mathbf{U} \in \mathcal{C}} \langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U} - \mathbf{U}^k \rangle + \frac{1}{2\eta} \|\mathbf{U} - \mathbf{U}^k\|_F^2 \quad (78)$$

$$= \mathcal{P}_{\mathcal{C}}(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k)), \quad (79)$$

For all $\mathbf{U} \in \mathcal{C}$, we have

$$\langle \nabla \hat{F}_p(\mathbf{U}^k) + \frac{1}{\eta}(\mathbf{U}^{k+1} - \mathbf{U}^k), \mathbf{U} - \mathbf{U}^{k+1} \rangle \geq 0. \quad (80)$$

Together with $\mathbf{U}^k \in \mathcal{C}$, we obtain

$$\langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U}^{k+1} - \mathbf{U}^k \rangle + \frac{1}{\eta} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 \leq 0. \quad (81)$$

Since $\hat{F}_p(\mathbf{U})$ is locally L_p -smooth for $\mathbf{U} \in \mathcal{C}$ as shown in Lemma 3 with $0 < \eta \leq 1/L_p$, we have

$$\begin{aligned} & \hat{F}_p(\mathbf{U}^{k+1}) \\ & \leq \hat{F}_p(\mathbf{U}^k) + \langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U}^{k+1} - \mathbf{U}^k \rangle + \frac{L_p}{2} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 \\ & \leq \hat{F}_p(\mathbf{U}^k) - \left(\frac{1}{\eta} - \frac{L_p}{2}\right) \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 \\ & \leq \hat{F}_p(\mathbf{U}^k) - \frac{L_p}{2} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2. \end{aligned} \quad (82)$$

It is obvious that $\hat{F}_p(\mathbf{U}^k) > 0$ and we know $\{\hat{F}_p(\mathbf{U}^k)\}$ is decreasing from (82), thus there exists some positive value $\hat{F}_p^*$ such that

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \hat{F}_p^*. \quad (83)$$

Besides, by summing the inequality (82) for $k = 0, 1, \dots, K-1$, we obtain

$$\begin{aligned} \frac{L_p}{2} \sum_{k=0}^{K-1} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F^2 & \leq \hat{F}_p(\mathbf{U}^0) - \hat{F}_p(\mathbf{U}^K) \\ & \leq \hat{F}_p(\mathbf{U}^0) - \hat{F}_p^* \leq \hat{F}_p(\mathbf{U}^0), \end{aligned} \quad (84)$$

where the last inequality uses $\hat{F}_p^* > 0$, and thus

$$\min_{k \in [K]} \|\mathbf{U}^{k+1} - \mathbf{U}^k\|_F \leq \sqrt{\frac{2\hat{F}_p(\mathbf{U}^0)}{L_p K}}. \quad (85)$$

APPENDIX B PROOF OF THEOREM 2

Before giving the result, we first review one important definition and one useful lemma.

Definition 2 (Gradient mapping). Suppose $\mathcal{Q} \subseteq \mathbb{C}^d$ is closed, convex, and nonempty, and f is differentiable. Given $t > 0$, the gradient mapping $G_t : \mathbb{C}^d \rightarrow \mathbb{C}^d$ is defined by

$$G_t(\mathbf{q}) = t(\mathbf{q} - \mathcal{P}_{\mathcal{Q}}(\mathbf{q} - \frac{1}{t} \nabla f(\mathbf{q}))), \quad \mathbf{q} \in \mathbb{R}^d, \quad (86)$$

where $\mathcal{P}_{\mathcal{Q}}(\cdot)$ denotes the Euclidean projection operator.

Lemma 4 ([31]). Denote $\bar{\mathbf{q}} = \mathcal{P}_{\mathcal{Q}}(\mathbf{q} - \eta \nabla f(\mathbf{q}))$, so that $G_{1/\eta}(\mathbf{q}) = \frac{1}{\eta}(\mathbf{q} - \bar{\mathbf{q}})$. If $\|G_{1/\eta}(\mathbf{q})\| \leq \varepsilon$ for some $\varepsilon \geq 0$, then

$$-\nabla f(\bar{\mathbf{q}}) \in N_{\mathcal{Q}}(\bar{\mathbf{q}}) + \mathcal{B}_2(0, \varepsilon(L\eta + 1)), \quad (87)$$

where L is the smoothness parameter of $f(\cdot)$.

Then we begin our proof. Define

$$G_{1/\eta}(\mathbf{U}^k) = \frac{1}{\eta}(\mathbf{U}^k - \mathcal{P}_{\mathcal{C}}(\mathbf{U}^k - \eta \nabla \hat{F}_p(\mathbf{U}^k))), \quad (88)$$

which means

$$G_{1/\eta}(\mathbf{U}^k) = \frac{1}{\eta}(\mathbf{U}^k - \mathbf{U}^{k+1}). \quad (89)$$

Since $1/\eta = L_p$, together with Theorem 1 we obtain

$$\min_{k \in [K]} \|G_{1/\eta}(\mathbf{U}^k)\|_F \leq \sqrt{\frac{2L_p \hat{F}_p(\mathbf{U}^0)}{K}}. \quad (90)$$

Since

$$K \geq \frac{8L\hat{F}_p(\mathbf{U}^0)}{\varepsilon^2}, \quad (91)$$

we get

$$\min_{k \in [K]} \|G_{1/\eta}(\mathbf{U}^k)\|_F \leq \frac{\varepsilon}{2}. \quad (92)$$

Together with Lemma 4, we complete the proof.

APPENDIX C PROOF OF THEOREM 3

Note that $\mathbf{U}^{k+1}$ is the solution to the following optimization problem:

$$\begin{aligned} \mathbf{U}^{k+1} & = \arg \min_{\mathbf{U} \in \mathcal{C}} \langle \nabla \hat{F}_p(\mathbf{U}^k), \mathbf{U} - \tilde{\mathbf{U}}^k \rangle + \frac{1}{2\eta} \|\mathbf{U} - \tilde{\mathbf{U}}^k\|_F^2 \\ & = \mathcal{P}_{\mathcal{C}}(\tilde{\mathbf{U}}^k - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k)), \end{aligned} \quad (93)$$

Then we will have the following descent result:

$$\begin{aligned} & \hat{F}_p(\mathbf{U}^{k+1}) \\ & \leq \hat{F}_p(\tilde{\mathbf{U}}^k) + \langle \nabla \hat{F}_p(\tilde{\mathbf{U}}^k), \mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k \rangle + \frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \\ & \leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \left(\frac{1}{\eta} - \frac{L_p}{2}\right) \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \\ & \leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2. \end{aligned} \quad (94)$$

The update rule of our algorithm guarantees that:

$$\hat{F}_p(\tilde{\mathbf{U}}^{k+1}) \leq \hat{F}_p(\mathbf{U}^{k+1}) \leq \hat{F}_p(\tilde{\mathbf{U}}^k) \leq \hat{F}_p(\mathbf{U}^k). \quad (95)$$

As $\hat{F}_p(\tilde{\mathbf{U}}^k), \hat{F}_p(\mathbf{U}^k) > 0$, we can conclude that

$$\lim_{k \rightarrow \infty} \hat{F}_p(\mathbf{U}^k) = \lim_{k \rightarrow \infty} \hat{F}_p(\tilde{\mathbf{U}}^k) = \hat{F}_p^\circ, \quad (96)$$

From (95), by induction we conclude that for all k

$$\hat{F}_p(\tilde{\mathbf{U}}^k) \leq \hat{F}_p(\mathbf{U}^0), \quad \hat{F}_p(\mathbf{U}^k) \leq \hat{F}_p(\mathbf{U}^0).$$

As $\mathbf{U}^k \in \mathcal{C}$, we have $\|\mathbf{U}^k\|_F \leq \sqrt{n} \|\mathbf{U}^k\|_{2,\infty} \leq \frac{2}{1+2\beta} \sqrt{\mu r \sigma}$. Besides,

$$\begin{aligned} \|\tilde{\mathbf{U}}^{k+1}\|_F & = \|(1+\beta)\mathbf{U}^{k+1} - \beta\mathbf{U}^k\|_F \\ & \leq (1+\beta)\|\mathbf{U}^{k+1}\|_F + \beta\|\mathbf{U}^k\|_F \leq 2\sqrt{\mu r \sigma}, \end{aligned}$$

where $\mathbf{U}^{k+1}, \mathbf{U}^k \in \mathcal{C}'$. By combining $\tilde{\mathbf{U}}^0 \in \mathcal{C}'$, we can conclude that $\{\tilde{\mathbf{U}}^k\}, \{\mathbf{U}^k\}$ are bounded and thus we have bounded limit points. Combining (94) and (95) yields

$$\begin{aligned} \frac{L_p}{2} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 & \leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \hat{F}_p(\mathbf{U}^{k+1}) \\ & \leq \hat{F}_p(\tilde{\mathbf{U}}^k) - \hat{F}_p(\tilde{\mathbf{U}}^{k+1}). \end{aligned}$$

By telescoping over k and letting $k \rightarrow \infty$, we obtain that

$$\frac{L_p}{2} \sum_{k=1}^{\infty} \|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \leq \hat{F}_p(\tilde{\mathbf{U}}^1) - \inf \hat{F}_p < \hat{F}_p(\tilde{\mathbf{U}}^1) < \infty,$$

and this implies that $\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F \rightarrow 0$ and hence that $\mathbf{U}^{k+1}$ and $\tilde{\mathbf{U}}^k$ share the same set of limit points Ω . As Ω is closed and bounded, we conclude that Ω is compact in $\mathbb{R}^d$.

Consider any limit point $\mathbf{U}' \in \Omega$, and w.l.o.g we write that $\mathbf{U}^{k+1} \rightarrow \mathbf{U}', \tilde{\mathbf{U}}^k \rightarrow \mathbf{U}'$ by restricting a subsequence. By the

optimality condition of the gradient updating step in (93), we have

$$-(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k) - \eta \nabla \hat{F}_p(\tilde{\mathbf{U}}^k) \in N_{\mathcal{C}}(\mathbf{U}^{k+1}).$$

Note that $\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k \rightarrow \mathbf{0}$ and $\mathbf{U}^{k+1}, \tilde{\mathbf{U}}^k \rightarrow \mathbf{U}'$, we conclude that $-\nabla \hat{F}_p(\mathbf{U}') \in N_{\mathcal{C}}(\mathbf{U}')$. By (96), we know $\hat{F}_p(\mathbf{U}') = \hat{F}_p^\circ$ for any $\mathbf{U}' \in \Omega$.

APPENDIX D PROOF OF THEOREM 4

From (93), we know

$$-\nabla \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{1}{\eta}(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k) \in N_{\mathcal{C}}(\mathbf{U}^{k+1}),$$

and it further implies that

$$\begin{aligned} \nabla \hat{F}_p(\mathbf{U}^{k+1}) - \nabla \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{1}{\eta}(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k) \\ \in \nabla \hat{F}_p(\mathbf{U}^{k+1}) + N_{\mathcal{C}}(\mathbf{U}^{k+1}). \end{aligned}$$

Let $\mathbf{E} = \nabla \hat{F}_p(\mathbf{U}^{k+1}) - \nabla \hat{F}_p(\tilde{\mathbf{U}}^k) - \frac{1}{\eta}(\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k)$. For $\tilde{\mathbf{U}}^k \in \mathcal{C}' \subseteq \mathcal{C}$, it is easy to verify that $\mathbf{U}^{k+1} \in \mathcal{C}$. As $\mathbf{U}^{k+1}, \tilde{\mathbf{U}}^k \in \mathcal{C}$ and $\hat{F}_p(\cdot)$ is locally L_p -Lipschitz from Lemma 3, we can obtain

$$\begin{aligned} \|\mathbf{E}\|_F &\leq \|\nabla \hat{F}_p(\mathbf{U}^{k+1}) - \nabla \hat{F}_p(\tilde{\mathbf{U}}^k)\|_F + \frac{1}{\eta}\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F \\ &\leq (L_p + \frac{1}{\eta})\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F, \end{aligned}$$

We know

$$\partial G(\mathbf{U}^{k+1}) = \nabla \hat{F}_p(\mathbf{U}^{k+1}) + N_{\mathcal{C}}(\mathbf{U}^{k+1}),$$

and we give the form of the distance as

$$\text{dist}_{\partial G(\mathbf{U})}(\mathbf{0}) = \min_{\mathbf{U} \in \partial G(\mathbf{U})} \|\mathbf{U}\|_F.$$

Combining (94) and (95), we know

$$\hat{F}_p(\mathbf{U}^{k+1}) \leq \hat{F}_p(\mathbf{U}^k) - \frac{L_p}{2}\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2.$$

As $\mathbf{E} \in \partial G(\mathbf{U}^{k+1})$, we have

$$\text{dist}_{\partial G(\mathbf{U}^{k+1})}(\mathbf{0}) \leq \|\mathbf{E}\|_F \leq (L_p + \frac{1}{\eta})\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F,$$

From the KL property of $G(\mathbf{U}^{k+1})$ and $G(\mathbf{U}^{k+1}) = \hat{F}_p(\mathbf{U}^{k+1})$ as $\mathbf{U}^{k+1} \in \mathcal{C}'$:

$$\varphi'(\hat{F}_p(\mathbf{U}^{k+1}) - \hat{F}_p^\circ) \text{dist}_{\partial G(\mathbf{U}^{k+1})}(\mathbf{0}) \geq \frac{1}{c_{\text{KL}}},$$

where $\varphi(t) = 2t^{\frac{1}{2}}$. Let $r_{k+1} = \hat{F}_p(\mathbf{U}^{k+1}) - \hat{F}_p^\circ$, we have

$$\begin{aligned} 1 &\leq (c_{\text{KL}}\varphi'(r_{k+1})\text{dist}_{\partial G(\mathbf{U}^{k+1})}(\mathbf{0}))^2 \\ &\leq c_{\text{KL}}^2(\varphi'(r_{k+1}))^2(L_p + \frac{1}{\eta})^2\|\mathbf{U}^{k+1} - \tilde{\mathbf{U}}^k\|_F^2 \\ &\leq c_{\text{KL}}^2(\varphi'(r_{k+1}))^2 \frac{2(L_p + \frac{1}{\eta})^2}{L_p}(\hat{F}_p(\mathbf{U}^k) - \hat{F}_p(\mathbf{U}^{k+1})) \\ &\leq c_{\text{KL}}^2 \frac{2(L_p + \frac{1}{\eta})^2}{L_p}(r_k - r_{k+1})(\varphi'(r_{k+1}))^2, \end{aligned}$$

and inserting $\varphi'(t) = \frac{1}{\sqrt{t}}$ and $\eta = \frac{1}{L_p}$, we establish that

$$r_{k+1} \leq \left(1 - \frac{1}{1 + C_p}\right)r_k,$$

where $C_p = 8c_{\text{KL}}^2 L_p$. Finally, we can prove Theorem 4 via the induction step.

APPENDIX E PROOF OF LEMMA 1

When $p = 1$, we get

$$\hat{f}_p(\mathbf{U}) = \|\phi_b(\mathbf{w})\|_1 = \langle \mathbf{1}, \phi_b(\mathbf{w}) \rangle, \quad (97)$$

then the differential of $\hat{f}_p(\mathbf{U})$ is

$$\begin{aligned} d\hat{f}_p(\mathbf{U}) &= \langle \mathbf{1}, \phi'_b(\mathbf{w}) \odot d\mathbf{w} \rangle = \langle \phi'_b(\mathbf{w}), d\mathbf{w} \rangle \\ &= \frac{1}{2} \langle \phi'_b(\mathbf{w}), \bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\mathbf{U}\mathbf{U}^T + \mathbf{U}d\mathbf{U}^T) \rangle \\ &\quad + \frac{1}{2} \langle \phi'_b(\mathbf{w}), \mathbf{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\bar{\mathbf{U}}\mathbf{U}^H + \bar{\mathbf{U}}d\mathbf{U}^H) \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y}), d\mathbf{U}\mathbf{U}^T \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}), d\bar{\mathbf{U}}\mathbf{U}^H \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}, d\mathbf{U} \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}})\mathbf{U}, d\bar{\mathbf{U}} \rangle \end{aligned} \quad (98)$$

where the second and fourth equalities apply

$$\langle \mathbf{A}, \mathbf{B} \odot \mathbf{D} \rangle = \langle \mathbf{A} \odot \bar{\mathbf{B}}, \mathbf{D} \rangle, \mathbf{A}, \mathbf{B}, \mathbf{D} \in \mathbb{C}^{m \times n}. \quad (99)$$

and the third equality uses

$$\begin{aligned} d\mathbf{w} &= \frac{1}{2}\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\mathbf{U}\mathbf{U}^T + \mathbf{U}d\mathbf{U}^T) \\ &\quad + \frac{1}{2}\mathbf{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\bar{\mathbf{U}}\mathbf{U}^H + \bar{\mathbf{U}}d\mathbf{U}^H). \end{aligned} \quad (100)$$

Therefore, we get

$$\nabla \hat{f}_p(\mathbf{U}) = \nabla_{\bar{\mathbf{U}}} \hat{f}_p(\mathbf{U}) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}. \quad (101)$$

When $p = 2$, using Facts 1 and 2 yields

$$\hat{f}_p(\mathbf{U}) = \frac{1}{2}\|\phi_b(\mathbf{w})\|_2^2 = \frac{1}{2}\langle \phi_b(\mathbf{w}), \phi_b(\mathbf{w}) \rangle, \quad (102)$$

then the differential of $\hat{f}_p(\mathbf{U})$ is

$$\begin{aligned} d\hat{f}_p(\mathbf{U}) &= \frac{1}{2} \langle \phi_b(\mathbf{w}), \phi'_b(\mathbf{w}) \odot d\mathbf{w} \rangle + \frac{1}{2} \langle \phi'_b(\mathbf{w}) \odot d\mathbf{w}, \phi_b(\mathbf{w}) \rangle \\ &= \langle \phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}), d\mathbf{w} \rangle \\ &= \frac{1}{2} \langle \phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}), \bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\mathbf{U}\mathbf{U}^T + \mathbf{U}d\mathbf{U}^T) \rangle \\ &\quad + \frac{1}{2} \langle \phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}), \mathbf{y} \odot \mathcal{D}^{-1} \mathcal{G}^*(d\bar{\mathbf{U}}\mathbf{U}^H + \bar{\mathbf{U}}d\mathbf{U}^H) \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}), d\mathbf{U}\mathbf{U}^T \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}), d\bar{\mathbf{U}}\mathbf{U}^H \rangle \\ &= \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{\mathbf{U}}, d\mathbf{U} \rangle \\ &\quad + \langle \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}})\mathbf{U}, d\bar{\mathbf{U}} \rangle, \end{aligned} \quad (103)$$

where the second equality applies $d\mathbf{w} = d\bar{\mathbf{w}}$, the third equality uses (100) and the fourth equality applies (99). Therefore, using Facts 1 and 2 obtains

$$\begin{aligned}\nabla \hat{f}_p(\mathbf{U}) &= \nabla_{\bar{\mathbf{U}}} \hat{f}_p(\mathbf{U}) \\ &= \mathcal{GD}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) \bar{\mathbf{U}}.\end{aligned}\quad (104)$$

APPENDIX F PROOF OF LEMMA 3

From Lemma 2 using triangle inequality yields

$$\|\nabla^2 \hat{f}_p(\mathbf{U})\| \leq 2(\|\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_p}\| + \|\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_p}\|). \quad (105)$$

Before giving the upper bound of $\|\nabla^2 \hat{f}_p(\mathbf{U})\|$, we first provide the upper bounds for other important components.

In terms of $\|\mathbf{\Gamma}_g\|$ and $\|\mathbf{\Gamma}_f\|$, noticing that $\sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T$ and $\mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T$ are projection matrices, we have

$$\left\| \mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \leq 1, \quad (106)$$

$$\left\| \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \leq 1. \quad (107)$$

Together with the fact that $\|\mathbf{K}_{s^2}\| = 1$, we have

$$\begin{aligned}\|\mathbf{\Gamma}_g\| &\leq (\|\mathbf{I}_{s^2}\| + \|\mathbf{K}_{s^2}\|) \left\| \mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \\ &\leq 2,\end{aligned}\quad (108)$$

$$\|\mathbf{\Gamma}_f\| \leq (\|\mathbf{I}_{s^2}\| + \|\mathbf{K}_{s^2}\|) \left\| \sum_{a=1}^n \text{vec}(\mathbf{H}_a) \text{vec}(\mathbf{H}_a)^T \right\| \leq 2. \quad (109)$$

As for $\|\mathbf{U}\|$, we obtain

$$\|\mathbf{U}\| \leq \|\mathbf{U}\|_F \leq \sqrt{n} \|\mathbf{U}\|_{2,\infty} \leq 2\sqrt{\mu r \sigma}. \quad (110)$$

Using the definition of $\phi_b(\cdot)$, we obtain

$$\|\phi'_b(\mathbf{w})\|_\infty \leq 1, \quad (111)$$

$$\|\phi''_b(\mathbf{w})\|_\infty \leq \frac{1}{2\sqrt{b}}, \quad (112)$$

$$\|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w})\|_\infty \leq 1/2, \quad (113)$$

$$\|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w})\|_\infty \leq \sqrt{\frac{8\mu^2 r^2 \sigma^2}{n^2} + \frac{b}{4}} + \frac{2\sqrt{2}\mu r \sigma}{n}, \quad (114)$$

where the last line applies the upper bound of $\|\mathbf{w}\|_\infty$ that

$$\begin{aligned}\|\mathbf{w}\|_\infty &= \|\text{Re}(\bar{\mathbf{y}} \odot \mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T))\|_\infty \\ &= \|\bar{\mathbf{y}}\|_\infty \|\mathcal{D}^{-1} \mathcal{G}^*(\mathbf{U}\mathbf{U}^T)\|_\infty \\ &\leq \sqrt{2} \|\mathbf{U}\mathbf{U}^T\|_\infty \leq \sqrt{2} \|\mathbf{U}\|_{2,\infty}^2 \leq \frac{4\sqrt{2}\mu r \sigma}{n},\end{aligned}$$

Together with $\|\mathcal{GD}^{-1}\| \leq 1$ and $\|\mathbf{y}\| = \sqrt{2n}$, we have

$$\|\mathcal{GD}^{-1}(\phi''_b(\mathbf{w}))\|_\infty \leq \|\phi''_b(\mathbf{w})\|_\infty \leq \frac{1}{2\sqrt{b}}, \quad (115)$$

$$\|\mathcal{GD}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\| \leq \|\phi'_b(\mathbf{w})\|_\infty \|\mathbf{y}\| \leq \sqrt{2n}, \quad (116)$$

$$\|\mathcal{GD}^{-1}(\phi''_b(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})\|_\infty \leq \|\phi''_b(\mathbf{w})\|_\infty \|\mathbf{y}\|_\infty^2 \leq \frac{1}{\sqrt{b}}, \quad (117)$$

$$\|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}\| \leq \|\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w})\|_\infty \|\mathbf{y}\| \quad (118)$$

When $p = 1$, we establish the upper bounds as follows:

$$\begin{aligned}\|\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_1}\| &\leq \|\mathbf{\Gamma}_f\| \|\mathbf{U}\|^2 \|\mathcal{GD}^{-1}(\phi''_b(\mathbf{w}))\|_\infty + \gamma \|\mathbf{\Gamma}_g\| \|\mathbf{U}\|^2, \\ &\leq 4\mu r \sigma \left(\frac{1}{\sqrt{b}} + 2\gamma \right),\end{aligned}\quad (119)$$

and

$$\begin{aligned}\|\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_1}\| &\leq \|\mathcal{GD}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\| \\ &\quad + \frac{1}{2} \|\mathbf{\Gamma}_f\| \|\mathbf{U}\|^2 \|\mathcal{GD}^{-1}(\phi''_b(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})\|_\infty + \gamma \|\mathbf{U}\|^2 \\ &\leq \sqrt{2n} + 4\mu r \sigma \left(\frac{1}{\sqrt{b}} + \gamma \right).\end{aligned}\quad (120)$$

Consequently, when $p = 1$, we get

$$L_p = 2\sqrt{2n} + 8\mu r \sigma \left(\frac{2}{\sqrt{b}} + 3\gamma \right). \quad (121)$$

Similarly, when $p = 2$, we establish the upper bounds as follows after careful calculation:

$$\begin{aligned}\|\mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{F}_2}\| &\leq \|\mathbf{\Gamma}_f\| \|\mathbf{U}\|^2 \|\mathcal{GD}^{-1}(\phi'_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) + \phi_b(\mathbf{w}) \odot \phi''_b(\mathbf{w}))\|_\infty \\ &\quad + \gamma \|\mathbf{\Gamma}_g\| \|\mathbf{U}\|^2 \\ &\leq 4\mu r \sigma (3 + 2\gamma),\end{aligned}\quad (122)$$

and

$$\begin{aligned}\|\mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{F}_2}\| &\leq \|\mathcal{GD}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\| + \gamma \|\mathbf{U}\|^2 \\ &\quad + \frac{1}{2} \|\mathbf{\Gamma}_f\| \|\mathbf{U}\|^2 \|\mathcal{GD}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})\|_\infty \\ &\leq (\sqrt{16\mu^2 r^2 \sigma^2 / n + nb/2} + 4\mu r \sigma / \sqrt{n}) + \frac{\mu r \sigma (3 + \gamma)}{2} \\ &\leq 4(\sqrt{\mu^2 r^2 \sigma^2 / n + nb} + \mu r \sigma / \sqrt{n}) + \frac{\mu r \sigma (3 + \gamma)}{2}.\end{aligned}$$

Consequently, when $p = 2$,

$$L_p = 8(\sqrt{\mu^2 r^2 \sigma^2 / n + nb} + \mu r \sigma / \sqrt{n}) + \mu r \sigma (9 + 3\gamma). \quad (123)$$

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Supplementary Material

APPENDIX G

NUMERICAL VERIFICATION OF KL PROPERTY FOR (73)

A key property of the KL constant is that a smaller $1/c_{\text{KL}}$ in (73) corresponds to a slower convergence rate of our algorithms. Therefore, we seek to verify whether our algorithms' $1/c_{\text{KL}}$ is as large as possible—a feature that directly indicates stable and efficient convergence. We present a plot of $\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0})$ corresponding to the left part in (73) to demonstrate the algorithms have a relatively large value of $1/c_{\text{KL}}$.

In the first experiment, we test $\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0})$ of the algorithm based on the ℓ_1 norm in the loss function, versus different iterations under different signal length $n = \{32, 40, 48, 56, 64\}$ and $\text{SNR} = \{-10, 0, 10, 20, 30, 40\}$ dB. In the second experiment, we apply the ℓ_2 norm in the loss function and test the algorithm. Other settings are the same as the first experiment.

From Fig 7 and Fig 8, we observe that $\varphi'(G(\mathbf{U}^k) - \hat{F}_p^\diamond) \text{dist}_{\partial G(\mathbf{U}^k)}(\mathbf{0})$ is lower bounded and thus there exists a constant $1/c_{\text{KL}}$ in (73) as large as possible to match this lower bound. Therefore, we finally verify that there exist a relatively large and stable KL constant $1/c_{\text{KL}}$ in our accelerated algorithms such that they exhibit stable and fast linear convergence both theoretically and empirically.

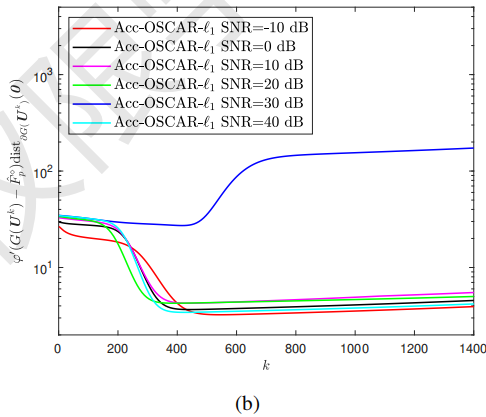
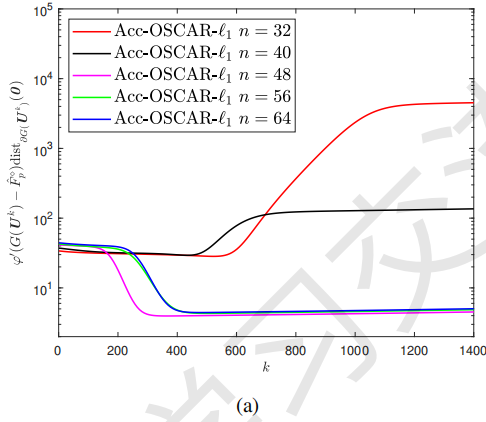


Fig. 7. KL constant v.s. k for Acc-OSCAR- ℓ_1 . (a) Different cases of $n = \{32, 40, 48, 56, 64\}$. (b) Different cases of $\text{SNR} = \{-10, 0, 10, 20, 30, 40\}$ dB.

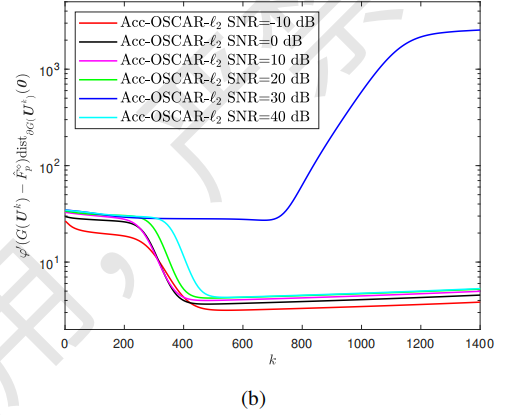
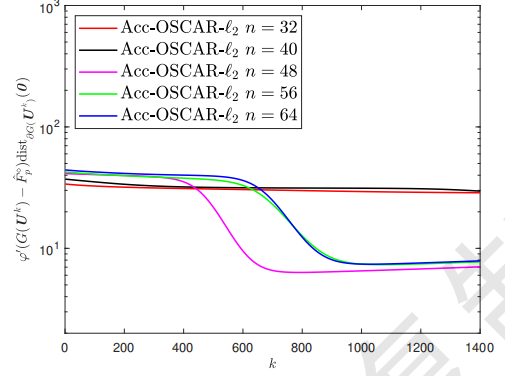


Fig. 8. KL constant v.s. k for Acc-OSCAR- ℓ_2 . (a) Different cases of $n = \{32, 40, 48, 56, 64\}$. (b) Different cases of $\text{SNR} = \{-10, 0, 10, 20, 30, 40\}$ dB.

APPENDIX H PROOF OF LEMMA 2

Before proving Lemma 2, we first give one useful lemma.

Lemma 5. For any $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, we have

$$\mathcal{G}(\mathbf{u}) \odot \mathcal{G}(\mathbf{v}) = \mathcal{G}\mathcal{D}^{-1}(\mathbf{u} \odot \mathbf{v}). \quad (124)$$

Proof. According to the definition of $\mathcal{H}$, we have

$$\mathcal{H}(\mathbf{u}') \odot \mathcal{H}(\mathbf{v}') = \mathcal{H}(\mathbf{u}' \odot \mathbf{v}'), \quad \mathbf{u}', \mathbf{v}' \in \mathbb{C}^n. \quad (125)$$

Define $\mathbf{u} = \mathcal{D}\mathbf{u}'$ and $\mathbf{v} = \mathcal{D}\mathbf{v}'$. We obtain

$$\mathcal{H}\mathcal{D}^{-1}(\mathbf{u}) \odot \mathcal{H}\mathcal{D}^{-1}(\mathbf{v}) = \mathcal{H}(\mathcal{D}^{-1}\mathbf{u} \odot \mathcal{D}^{-1}\mathbf{v}) \quad (126)$$

$$= \mathcal{H}\mathcal{D}^{-2}(\mathbf{u} \odot \mathbf{v}). \quad (127)$$

Using the fact $\mathcal{G} = \mathcal{H}\mathcal{D}^{-1}$ completes the proof. $\square$

Then we start to prove Lemma 2. Based on (27), the Hessian matrices $\nabla^2 \hat{F}_p(\mathbf{U})$ are

$$\nabla^2 \hat{F}_p(\mathbf{U}) = \nabla^2 \hat{f}_p(\mathbf{U}) + \gamma \nabla^2 g(\mathbf{U}), \quad p = 1, 2. \quad (128)$$

Here, $\nabla^2 \hat{f}_p(\mathbf{U})$ and $\nabla^2 g(\mathbf{U})$ are the Hessian matrices of $\hat{f}_p(\mathbf{U})$ and $g(\mathbf{U})$, defined by

$$\nabla^2 \hat{f}_p(\mathbf{U}) = \begin{bmatrix} \mathbf{H}_{\bar{\mathbf{U}}, \bar{\mathbf{U}}}^{\hat{f}_p} & \mathbf{H}_{\bar{\mathbf{U}}, \mathbf{U}}^{\hat{f}_p} \\ \mathbf{H}_{\mathbf{U}, \bar{\mathbf{U}}}^{\hat{f}_p} & \mathbf{H}_{\mathbf{U}, \mathbf{U}}^{\hat{f}_p} \end{bmatrix}, \quad (129)$$

$$\nabla^2 g(U) = \begin{bmatrix} \mathbf{H}_{\bar{U},U}^g & \mathbf{H}_{\bar{U},\bar{U}}^g \\ \mathbf{H}_{U,U}^g & \mathbf{H}_{U,\bar{U}}^g \end{bmatrix}, \quad (130)$$

where $\mathbf{H}_{\bar{U},U}^{\hat{f}_p}$, $\mathbf{H}_{\bar{U},\bar{U}}^{\hat{f}_p}$, $\mathbf{H}_{U,U}^{\hat{f}_p}$ and $\mathbf{H}_{U,\bar{U}}^{\hat{f}_p}$ are the complex Hessian matrices of $\hat{f}_p(U)$ and $\mathbf{H}_{\bar{U},U}^g$, $\mathbf{H}_{\bar{U},\bar{U}}^g$, $\mathbf{H}_{U,U}^g$ and $\mathbf{H}_{U,\bar{U}}^g$ are the complex Hessian matrices of $g(\bar{U})$ in Fact 3.

Since $\nabla^2 g(U)$ appears in both $p = 1$ and $p = 2$, we consider $\nabla^2 g(U)$ first. As shown in (37), we have provided the gradient of $\nabla g(U)$. The differential of $\nabla_{\bar{U}} g(U) = \nabla g(U)$ is

$$\begin{aligned} d\nabla_{\bar{U}} g(U) &= (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(dUU^T)\bar{U} + (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UdU^T)\bar{U} \\ &\quad + (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T)d\bar{U}. \end{aligned} \quad (131)$$

Define

$$\mathbf{M}_1 = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(dUU^T)\bar{U}, \quad (132)$$

$$\mathbf{M}_2 = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UdU^T)\bar{U}, \quad (133)$$

$$\mathbf{M}_3 = (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T)d\bar{U}. \quad (134)$$

Then using (49), we have

$$\mathbf{M}_1 = dUU^T\bar{U} - \sum_{a=1}^n \langle \mathbf{H}_a, dUU^T \rangle \mathbf{H}_a \bar{U}, \quad (135)$$

$$\mathbf{M}_2 = UdU^T\bar{U} - \sum_{a=1}^n \langle \mathbf{H}_a, UdU^T \rangle \mathbf{H}_a \bar{U} \quad (136)$$

Recalling the property of Kronecker product, for $\mathbf{A} \in \mathbb{C}^{n_1 \times n_2}$, $\mathbf{X} \in \mathbb{C}^{n_2 \times n_3}$, $\mathbf{B} \in \mathbb{C}^{n_3 \times n_4}$, we know

$$\text{vec}(\mathbf{AXB}) = (\mathbf{B}^T \otimes \mathbf{A})\text{vec}(\mathbf{X}). \quad (137)$$

Then for $\mathbf{M}_1$, we have

$$\begin{aligned} \text{vec}(\mathbf{M}_1) &= (\mathbf{U}^H \mathbf{U} \otimes \mathbf{I}_s)\text{vec}(dU) - (\mathbf{U}^H \otimes \mathbf{I}_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(\mathbf{H}_a)\text{vec}(\mathbf{H}_a)^T \right] \text{vec}(dUU^T) \\ &= (\mathbf{U}^H \mathbf{U} \otimes \mathbf{I}_s)\text{vec}(dU) - (\mathbf{U}^H \otimes \mathbf{I}_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(\mathbf{H}_a)\text{vec}(\mathbf{H}_a)^T \right] (\mathbf{U} \otimes \mathbf{I}_s)\text{vec}(dU). \end{aligned} \quad (138)$$

For $\mathbf{M}_2$, we have

$$\begin{aligned} \text{vec}(\mathbf{M}_2) &= (\mathbf{U}^H \otimes \mathbf{I}_s)\text{vec}(UdU^T) - (\mathbf{U}^H \otimes \mathbf{I}_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(\mathbf{H}_a)\text{vec}(\mathbf{H}_a)^T \right] \text{vec}(UdU^T) \\ &= (\mathbf{U}^H \otimes \mathbf{I}_s)\mathbf{K}_{s^2}(U \otimes \mathbf{I}_s)\text{vec}(dU) - (\mathbf{U}^H \otimes \mathbf{I}_s) \\ &\quad \cdot \left[\sum_{a=1}^n \text{vec}(\mathbf{H}_a)\text{vec}(\mathbf{H}_a)^T \right] \mathbf{K}_{s^2}(U \otimes \mathbf{I}_s)\text{vec}(dU). \end{aligned} \quad (139)$$

For $\mathbf{M}_3$, we have

$$\text{vec}(\mathbf{M}_3) = (\mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T))\text{vec}(d\bar{U}). \quad (140)$$

Therefore,

$$\text{vec}(d\nabla g(U)) = (\mathbf{U}^H \otimes \mathbf{I}_s) \left(\mathbf{I}_{s^2} - \sum_{a=1}^n \text{vec}(\mathbf{H}_a)\text{vec}(\mathbf{H}_a)^T \right)$$

$$\begin{aligned} &\cdot (\mathbf{I}_{s^2} + \mathbf{K}_{s^2})(U \otimes \mathbf{I}_s)\text{vec}(dU) \\ &\quad + (\mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T))\text{vec}(d\bar{U}) \\ &= (\mathbf{U}^H \otimes \mathbf{I}_s)\mathbf{\Gamma}_g(U \otimes \mathbf{I}_s)\text{vec}(dU) \\ &\quad + (\mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T))\text{vec}(d\bar{U}). \end{aligned} \quad (141)$$

As for $\nabla_U g(U) = \overline{\nabla_{\bar{U}} g(U)}$, we have

$$\begin{aligned} \text{vec}(d\nabla_U g(U)) &= \overline{\text{vec}(d\nabla_{\bar{U}} g(U))} \\ &= (\mathbf{U}^T \otimes \mathbf{I}_s)\mathbf{\Gamma}_g(\bar{U} \otimes \mathbf{I}_s)\text{vec}(d\bar{U}) \\ &\quad + (\mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\bar{U}\bar{U}^H))\text{vec}(dU). \end{aligned} \quad (142)$$

Then we obtain

$$\begin{aligned} d^2 g(U) &= \left(\text{vec}(d\nabla_U g(U)) \right)^T \text{vec}(dU) \\ &\quad + \left(\text{vec}(d\nabla_{\bar{U}} g(U)) \right)^T \text{vec}(d\bar{U}). \end{aligned} \quad (143)$$

Combining (141), (142) and (143) gives

$$\mathbf{H}_{\bar{U},U}^g = (\mathbf{U}^T \otimes \mathbf{I}_s)\mathbf{\Gamma}_g(\bar{U} \otimes \mathbf{I}_s), \quad (144)$$

$$\mathbf{H}_{U,\bar{U}}^g = (\mathbf{U}^H \otimes \mathbf{I}_s)\mathbf{\Gamma}_g(U \otimes \mathbf{I}_s), \quad (145)$$

$$\mathbf{H}_{U,U}^g = \mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(\bar{U}\bar{U}^H), \quad (146)$$

$$\mathbf{H}_{\bar{U},\bar{U}}^g = \mathbf{I}_r \otimes (\mathcal{I} - \mathcal{G}\mathcal{G}^*)(UU^T). \quad (147)$$

Next, we consider the Hessian matrices of $\hat{f}_p(U)$ for $p = 1$ and $p = 2$, respectively. When $p = 1$, according to Eq. (98), we get

$$\nabla_{\bar{U}} \hat{f}_1(U) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{U}, \quad (148)$$

$$\nabla_U \hat{f}_1(U) = \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}})U. \quad (149)$$

Together with Lemma 5 after careful calculation, we obtain

$$\mathbf{H}_{\bar{U},U}^{\hat{f}_1} = (\mathbf{U}^T \otimes \mathbf{I}_s)\mathbf{\Gamma}_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}))) (\bar{U} \otimes \mathbf{I}_s), \quad (150)$$

$$\mathbf{H}_{U,\bar{U}}^{\hat{f}_1} = (\mathbf{U}^H \otimes \mathbf{I}_s)\mathbf{\Gamma}_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}))) (U \otimes \mathbf{I}_s), \quad (151)$$

$$\begin{aligned} \mathbf{H}_{U,U}^{\hat{f}_1} &= \mathbf{I}_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}) + \frac{1}{2}(\mathbf{U}^T \otimes \mathbf{I}_s)\mathbf{\Gamma}_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}) \odot \bar{\mathbf{y}} \odot \bar{\mathbf{y}})) (U \otimes \mathbf{I}_s), \end{aligned} \quad (152)$$

$$\begin{aligned} \mathbf{H}_{\bar{U},\bar{U}}^{\hat{f}_1} &= \mathbf{I}_r \otimes \mathcal{G}\mathcal{D}^{-1}(\phi'_b(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2}(\mathbf{U}^H \otimes \mathbf{I}_s)\mathbf{\Gamma}_f \\ &\quad \cdot \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\phi''_b(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y})) (\bar{U} \otimes \mathbf{I}_s). \end{aligned} \quad (153)$$

When $p = 2$, we have

$$\nabla_{\bar{U}} \hat{f}_2(U) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y})\bar{U}, \quad (154)$$

$$\nabla_U \hat{f}_2(U) = \mathcal{G}\mathcal{D}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}})U. \quad (155)$$

After careful calculation, we obtain

$$\mathbf{H}_{\bar{U},U}^{\hat{f}_2} = (\mathbf{U}^T \otimes \mathbf{I}_s)\mathbf{\Gamma}_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}))) (\bar{U} \otimes \mathbf{I}_s), \quad (156)$$

$$\mathbf{H}_{U,\bar{U}}^{\hat{f}_2} = (\mathbf{U}^H \otimes \mathbf{I}_s)\mathbf{\Gamma}_f \text{diag}(\mathcal{G}\mathcal{D}^{-1}(\Lambda_\phi(\mathbf{w}))) (U \otimes \mathbf{I}_s), \quad (157)$$

$$\begin{aligned} \mathbf{H}_{U,U}^{\hat{f}_2} = & \mathbf{I}_r \otimes \mathcal{GD}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \bar{\mathbf{y}}) + \frac{1}{2}(\mathbf{U}^T \otimes \mathbf{I}_s) \mathbf{\Gamma}_f \\ & \cdot \text{diag} \left(\mathcal{GD}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \bar{\mathbf{y}} \odot \bar{\mathbf{y}}) \right) (\mathbf{U} \otimes \mathbf{I}_s), \quad (158) \end{aligned}$$

$$\begin{aligned} \mathbf{H}_{\bar{U},\bar{U}}^{\hat{f}_2} = & \mathbf{I}_r \otimes \mathcal{GD}^{-1}(\phi_b(\mathbf{w}) \odot \phi'_b(\mathbf{w}) \odot \mathbf{y}) + \frac{1}{2}(\mathbf{U}^H \otimes \mathbf{I}_s) \mathbf{\Gamma}_f \\ & \cdot \text{diag} \left(\mathcal{GD}^{-1}(\Lambda_\phi(\mathbf{w}) \odot \mathbf{y} \odot \mathbf{y}) \right) (\bar{\mathbf{U}} \otimes \mathbf{I}_s). \quad (159) \end{aligned}$$

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